

Virtual Try-On — Final Report

Magic Hour take-home. Seven hosted try-on arms, a staged budget-gated evaluation, and a shipped cascade — every number below is computed from `cv_metrics.csv` and `vlm_judgments.csv`.

10

arms evaluated
(7 models + cascade)

30

unique person × garment pairs

250

scored generations

250

blind VLM judgments

SECTION 01

Approach

Goal: a Colab that takes a person image + a garment image and produces the person wearing that garment, demonstrably better than the website's current model (**Qwen 2511**). Seven hosted candidate models ("arms") were run through a staged, budget-gated evaluation pipeline, judged by three independent systems:

- **Deterministic evaluator harness** (authoritative for elimination): garment color-fidelity (HSV histogram, torso crop vs reference), identity (ArcFace face-embedding cosine), pose consistency (MediaPipe landmark displacement), and background preservation (PSNR outside the person mask). Scores combine into a weighted composite (garment ×2) normalized on fixed absolute anchors.
- **VLM judge** (gpt-5.5, blind to model names): a six-criterion rubric — garment, identity, scene, clean, hands, realism — schema-validated with self-correcting retries.
- **Human review** as the supervising tiebreaker wherever the judges disagreed.

Status note: the deterministic harness is a work in progress — table values are produced under human supervision (rankings validated, flagged judge disagreements adjudicated by human review).

Stages: **triage** (7 arms × 4 pairs) → automatic elimination (top 50% + baseline) → **grid** (5 arms × 12 pairs, plus a best-of-2 seed-stability pass on the top 2) → **held-out benchmark** (18 pairs never seen by any selection decision). Every generation is a reproducible run package; a top-to-bottom re-run reproduces the same elimination and spends nothing on work already done.

SECTION 02

Deliverables

- **The Key** — frozen serving configuration. Default implementation: `seedream5_lite` → `qwen_image3` `realism-refine cascade` ; single-model `seedream5_lite` ships alongside it.
- **MVP eval harness** — model-agnostic (one registry entry per new model), stage-agnostic, with deterministic + VLM + human layers; reusable for future model bake-offs.
- **Comparison module** — `compare_vs_baseline()` runs cascade, single seedream, and the Magic Hour baseline on the same inputs, side by side.
- **Evidence** — all boards below, plus per-output scores in `cv_metrics.csv` / `vlm_judgments.csv` and this report's side-by-side gallery.

Why the cascade: complementary strengths (the checkbox analysis)

The VLM rubric acts as per-model checkboxes. Aggregating them into two axes — **editing** (garment + identity + scene) and **realism** (clean + hands + realism) — showed seedream as the strongest transferer while `qwen_image3` rendered realistically but edited poorly. The cascade built from that observation was then measured directly on the grid stage: `best editing axis (4.33)`, `best garment score of any arm (4.00)`, `identity 4.55 vs plain seedream's 4.33` per the blind VLM. The deterministic pixel-composite did not favor it (re-rendering trades pixel preservation for perceptual quality); the final selection is human review corroborated by the frontier VLM. The baseline was beaten by every judging system on unseen data.

SECTION 03

Results by stage

Click a column header to sort. Composite / overall cells are shaded by value (darker green = higher). The top row of each board on its default metric is highlighted.

Holdout

18 unseen pairs — the reported numbers. No selection decision ever saw these pairs.

Deterministic composite (fixed anchors, garment ×2 — higher is better; pose_err lower is better)

arm	garment_sim	identity_cos	pose_err	bg_psnr	score
flux_vto_v1 BEST	0.108	0.875	0.058	28.57	0.642
fashn_v16	0.097	0.974	0.046	29.19	0.632
seedream_qwen_refine (cascade)	0.129	0.730	0.104	20.36	0.522
seedream5_lite	0.106	0.754	0.101	20.98	0.507
qwen_2511 (baseline)	0.077	0.855	0.241	16.16	0.385

VLM judge (gpt-5.5, blind to model names; 1–5 per criterion)

arm	garment	identity	scene	clean	hands	realism
seedream_qwen_refine (cascade) BEST	3.94	4.50	4.67	4.06	3.94	4.06
seedream5_lite	3.89	4.44	4.67	4.00	3.78	4.00
flux_vto_v1	3.11	4.61	4.83	3.89	3.83	4.00
qwen_2511 (baseline)	3.39	4.00	4.28	3.61	3.50	3.72
fashn_v16	2.89	4.44	4.61	3.39	3.17	3.61

Complementarity axes (editing = garment+identity+scene; realism = clean+hands+realism)

arm	editing	realism	edit_rank	realism_rank
seedream_qwen_refine (cascade) BEST	4.37	4.02	1	1
seedream5_lite	4.33	3.93	2	2
flux_vto_v1	4.19	3.91	3	3
fashn_v16	3.98	3.39	4	5
qwen_2511 (baseline)	3.89	3.61	5	4

Grid

12 pairs across the surviving arms; includes the cascade arm and a best-of-2 seed-stability pass on the top 2.

Deterministic composite (fixed anchors, garment x2 — higher is better; pose_err lower is better)

arm	garment_sim	identity_cos	pose_err	bg_psnr	score
flux_vto_v1 BEST	0.122	0.872	0.025	28.63	0.677
fashn_v16	0.127	0.980	0.034	28.49	0.666
klein_4b_edit	0.105	0.908	0.029	25.48	0.631
composite_v2	0.141	0.773	0.044	22.63	0.586
seedream5_lite	0.200	0.686	0.067	20.69	0.569
magic_hour_api (website)	0.145	0.647	0.034	25.90	0.557
seedream_qwen_refine (cascade)	0.142	0.660	0.073	20.59	0.507
qwen_2511 (baseline)	0.130	0.814	0.188	15.01	0.404

VLM judge (gpt-5.5, blind to model names; 1–5 per criterion)

arm	garment	identity	scene	clean	hands	realism	overall
composite_v2 BEST	2.50	4.83	4.75	4.42	4.50	4.50	4.25
flux_vto_v1	3.21	4.71	4.96	3.92	3.96	4.04	4.13
klein_4b_edit	3.42	4.58	5.00	3.92	3.83	4.00	4.12
seedream_qwen_refine (cascade)	4.00	4.25	4.42	4.00	3.83	4.08	4.10
seedream5_lite	4.00	4.33	4.38	3.96	3.88	4.04	4.10
magic_hour_api (website)	3.08	4.25	4.58	3.58	3.83	3.75	3.85
qwen_2511 (baseline)	3.42	3.83	4.33	3.67	3.92	3.83	3.83
fashn_v16	3.38	4.38	4.71	3.50	3.38	3.58	3.82

Complementarity axes (editing = garment+identity+scene; realism = clean+hands+realism)

arm	editing	realism	edit_rank	realism_rank
klein_4b_edit BEST	4.33	3.92	1	5
flux_vto_v1	4.29	3.97	2	2
seedream5_lite	4.24	3.96	3	4

arm	editing	realism	edit_rank	realism_rank
seedream_qwen_refine (cascade)	4.22	3.97	4	2
fashn_v16	4.15	3.49	5	8
composite_v2	4.03	4.47	6	1
magic_hour_api (website)	3.97	3.72	7	7
qwen_2511 (baseline)	3.86	3.81	8	6

Triage

All 7 candidate arms on 4 pairs; bottom half eliminated on the deterministic composite.

Deterministic composite (fixed anchors, garment x2 — higher is better; pose_err lower is better)

arm	garment_sim	identity_cos	pose_err	bg_psnr	score
fashn_v16 BEST	0.181	0.971	0.024	28.90	0.728
flux_vto_v1	0.092	0.797	0.029	28.10	0.622
klein_4b_edit	0.065	0.829	0.026	19.73	0.606
seedream5_lite	0.247	0.410	0.052	22.18	0.523
qwen_2511 (baseline)	0.149	0.851	0.128	16.07	0.508
qwen_image3_edit	0.147	0.798	0.235	17.03	0.490
klein_tryon_lora	0.159	0.558	0.164	17.25	0.454

VLM judge (gpt-5.5, blind to model names; 1–5 per criterion)

arm	garment	identity	scene	clean	hands	realism	overall
seedream5_lite BEST	4.00	4.50	5.00	4.00	4.00	4.00	4.25
fashn_v16	4.00	4.75	4.75	3.75	3.75	3.75	4.12
flux_vto_v1	3.25	4.25	5.00	4.00	4.00	4.00	4.08
qwen_image3_edit	3.75	4.00	4.50	3.75	3.50	4.00	3.92
qwen_2511 (baseline)	3.25	4.25	4.75	3.25	3.75	3.25	3.75

arm	garment	identity	scene	clean	hands	realism	overall
klein_4b_edit	3.25	3.75	4.00	3.25	4.25	3.25	3.62
klein_tryon_lora	2.50	3.00	2.75	4.00	3.75	4.00	3.33

Complementarity axes (editing = garment+identity+scene; realism = clean+hands+realism)

arm	editing	realism	edit_rank	realism_rank
seedream5_lite BEST	4.50	4.00	1	1
fashn_v16	4.50	3.75	1	4
flux_vto_v1	4.17	4.00	3	1
qwen_image3_edit	4.08	3.75	4	4
qwen_2511 (baseline)	4.08	3.42	4	7
klein_4b_edit	3.67	3.58	6	6
klein_tryon_lora	2.75	3.92	7	3

SECTION 04

Side-by-side gallery

Identical inputs across four configurations: raw qwen_2511 at defaults, the Magic Hour production output (downloaded straight from the Magic Hour AI Clothes Changer API, unmodified, native 384x576), and our two implementations. A column is omitted where that output does not exist (the website API ran on the 12 grid pairs). Under each output: its deterministic component scores and composite for that stage/arm/pair, plus the blind VLM overall where judged. Click any image to enlarge (arrow keys to step, Esc or click to close).

GRID

p001xg015



person



garment



raw qwen_2511 (defaults)
det 0.43 · g 0.07 · id 0.80 · pose
0.146 · psnr 18.2
VLM 3.83 / 5



Magic Hour
det 0.40 · g 0.028
VLM

GRID p002xg022



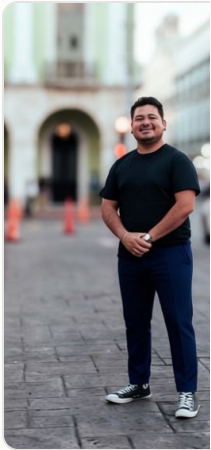
person



garment



raw qwen_2511 (defaults)
det 0.31 · g -0.01 · id 0.90 · pose
0.146 · psnr 14.3
VLM 4.17 / 5



Magic Hour
det 0.48 · g -0.003
VLM 4.1

GRID p005xg023



person



garment



raw qwen_2511 (defaults)
det 0.37 · g 0.09 · id 0.90 · pose
0.170 · psnr 12.3
VLM 4.00 / 5



Magic Hour well
outp
det 0.82 · g 0.47
0.024 · ps
VLM 4.

GRID p006xg006



person



garment



raw qwen_2511 (defaults)
det 0.40 · g 0.07 · id 0.81 · pose
0.142 · psnr 15.2
VLM 4.17 / 5



Magic Hour well
outp
det 0.48 · g 0.13
0.014 · ps
VLM 3.

GRID p007xg019



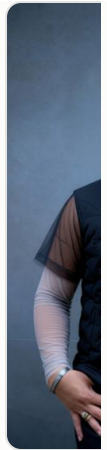
person



garment



raw qwen_2511 (defaults)
det 0.40 · g -0.01 · id 0.97 · pose
0.052 · psnr 15.9
VLM 3.83 / 5



Magic

det 0.5

GRID p008xg024



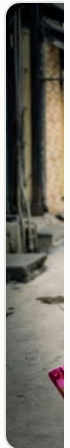
person



garment



raw qwen_2511 (defaults)
det 0.03 · g 0.01 · id 0.04 · pose
0.750 · psnr 13.9
VLM 2.33 / 5



Mi

det

GRID p013xg004



person



garment



raw q
det 0.2!
(



GRID p003xg011



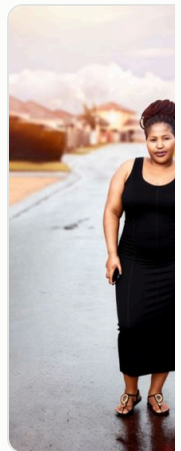
person



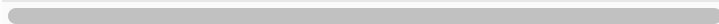
garment



raw qwen_2511 (defaults)
det 0.69 · g 0.25 · id 0.87 · pose
0.077 · psnr 18.9
VLM 3.67 / 5



Magic Hour
c
det 0.57 · g C
0.021
VLM



GRID p009xg002



person



garment



raw qwen_2511 (defaults)
det 0.34 · g 0.02 · id 0.98 · pose
0.118 · psnr 13.8
VLM 4.00 / 5



Magic Hour well
outp
det 0.69 · g 0.12
0.041 · ps
VLM 4.

GRID p010xg001



person



garment



raw qwen_2511 (defaults)
det 0.73 · g 0.81 · id 0.84 · pose
0.123 · psnr 15.2
VLM 4.17 / 5



Magic Hour well
outp
det 0.77 · g 0.73
0.020 · ps
VLM 4.

GRID p011xg028



person



garment



raw qwen_2511 (defaults)
det 0.60 · g 0.19 · id 0.98 · pose
0.076 · psnr 15.8
VLM 4.00 / 5



Magic Hour website
output)
det 0.60 · g 0.19 · id 0.98 · pose
0.022 · psnr 15.8
VLM

GRID p012xg007



person



garment



raw qwen_2511 (defaults)
det 0.29 · g 0.01 · id 0.91 · pose
0.176 · psnr 14.0
VLM 4.17 / 5



Magic Hour website
output)
det 0.39 · g 0.04 · id 0.91 · pose
0.104 · psnr 14.0
VLM 3.33 / 5

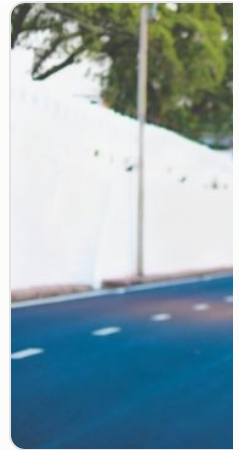
HOLDOUT p004xg010



person



garment



raw
det 0.

HOLDOUT

p014xg008



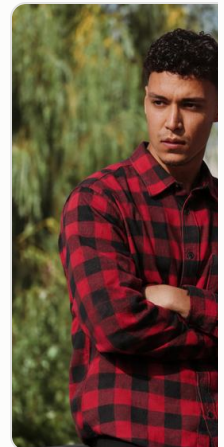
person



garment



raw qwen_2511 (defaults)
det 0.36 · g 0.01 · id 0.97 · pose
0.085 · psnr 14.4
VLM 4.33 / 5



seedream (o
det 0.56 · g 0.08
0.052 · ps
VLM 4.

HOLDOUT

p015xg026



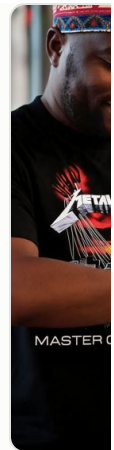
person



garment



raw qwen_2511 (defaults)
det 0.36 · g 0.02 · id 0.98 · pose
0.086 · psnr 13.0
VLM 3.50 / 5



see
det 0.4



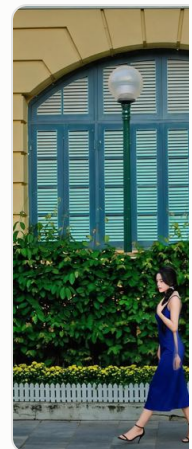
person



garment



raw qwen_2511 (defaults)
det 0.03 · g 0.02 · id -0.03 · pose
2.137 · psnr 12.8
VLM 2.50 / 5



seedream
det 0.06 · g 0.
0.384
VLM

HOLDOUT p017xg029



person



garment



raw qwen_2511 (defaults)
det 0.46 · g 0.09 · id 0.95 · pose
0.105 · psnr 16.5
VLM 3.83 / 5



seedream (ours)
det 0.79 · g 0.26 ·
0.034 · psi
VLM 4.3

HOLDOUT p018xg027



person



garment



raw qwen_2511 (defaults)
det 0.35 · g 0.02 · id 0.90 · pose
0.139 · psnr 15.9
VLM 3.33 / 5

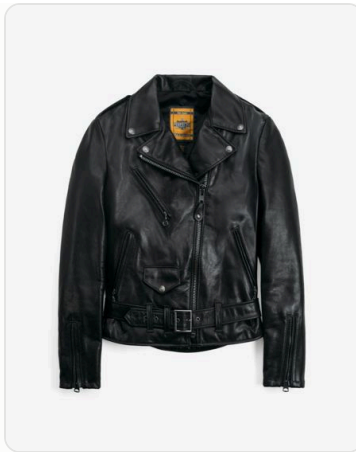


seedream (ours, sin)
det 0.19 · g -0.01 · id 0.7
0.277 · psnr 13.2
VLM 4.00 / 5

HOLDOUT p019xg017



person



garment



raw
det 0.4

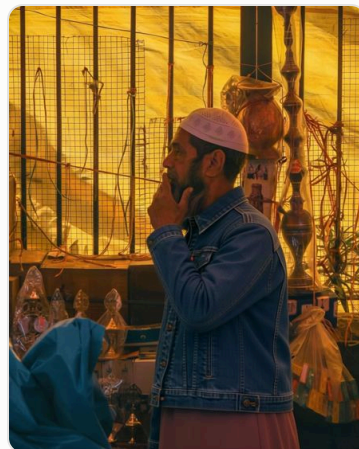
HOLDOUT p020xg016



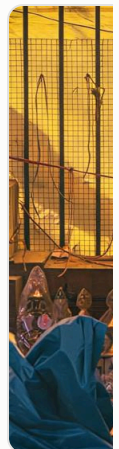
person



garment



raw qwen_2511 (defaults)
det nan · g -0.01 · id 0.95 · pose nan
· psnr 14.7
VLM 4.33 / 5



see
det nan ·

HOLDOUT p021xg012



person



garment



raw qwen_2511 (defaults)
det 0.42 · g 0.10 · id 0.94 · pose
0.165 · psnr 16.0
VLM 3.67 / 5



seedream
det 0.58 · g 0.097
VLM

HOLDOUT

p022xg021



person



garment



raw qwen_2511 (defaults)
det 0.30 · g -0.01 · id 0.81 · pose
0.153 · psnr 14.7
VLM 4.00 / 5



seedream (o
det 0.39 · g 0.00
0.057 · ps
VLM 4.

HOLDOUT

p023xg018



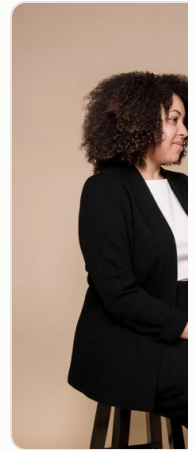
person



garment



raw qwen_2511 (defaults)
det 0.38 · g -0.02 · id 0.94 · pose
0.087 · psnr 16.6
VLM 3.67 / 5



seedream
det 0.44 · g 0.
0.176
VLM

HOLDOUT

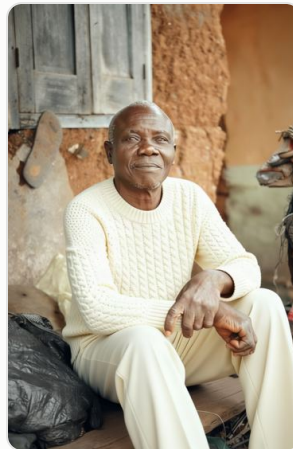
p024xg009



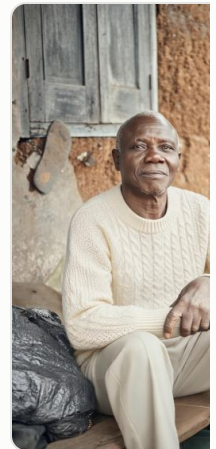
person



garment



raw qwen_2511 (defaults)
det 0.72 · g 0.93 · id 0.97 · pose
0.132 · psnr 14.5
VLM 4.00 / 5



seedream (c
det 0.70 · g 0.30
0.152 · p
VLM 3

HOLDOUT

p025xg013



person



garment



raw qwen_2511 (defaults)
det 0.23 · g 0.04 · id 0.71 · pose
0.262 · psnr 14.7
VLM 4.17 / 5



seedrea
det 0.19 · g
0.15
VL

HOLDOUT p026xg020



person



garment

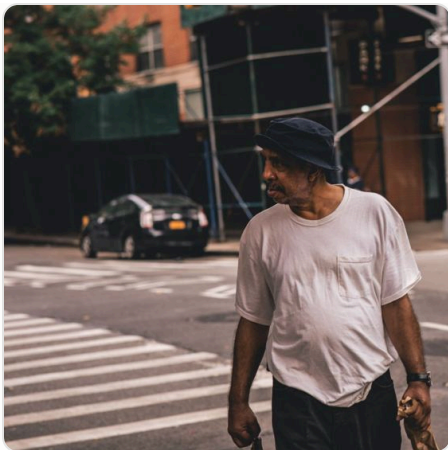


raw qwen_2511 (defaults)
det 0.27 · g 0.02 · id 0.82 · pose
0.190 · psnr 10.0
VLM 4.33 / 5



seedream
det 0.40 · g 0.
0.045
VLM

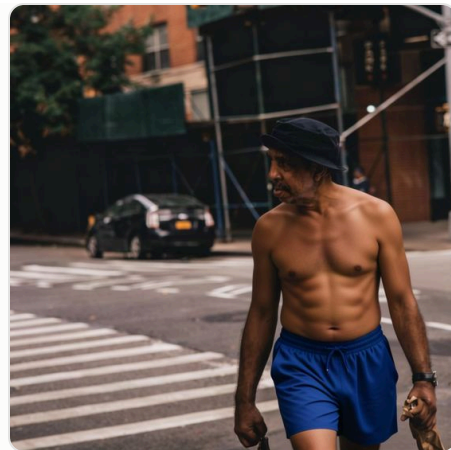
HOLDOUT p027xg025



person



garment



raw qwen_2511 (defaults)
det 0.55 · g -0.01 · id 0.95 · pose
0.038 · psnr 30.5
VLM 3.00 / 5

HOLDOUT

p028xg030



person



garment

HOLDOUT

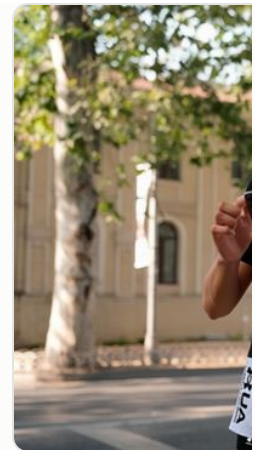
p029xg005



person



garment



raw
det 0.3



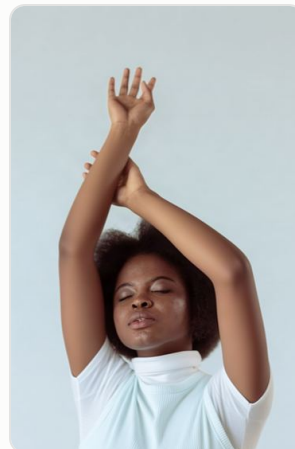
HOLDOUT p030xg003



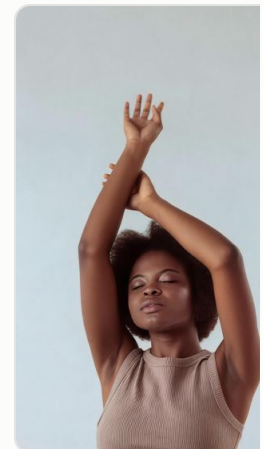
person



garment



raw qwen_2511 (defaults)
det 0.36 · g 0.01 · id 0.97 · pose
0.133 · psnr 17.9
VLM 3.00 / 5



seedream (ours)
det 0.69 · g 0.17 · id
0.061 · psnr
VLM 4.17



Generated from runs/cv_metrics.csv and runs/vlm_judgments.csv. Deterministic composite: per-metric normalization on fixed anchors (garment_sim 0→0.35, identity_cos 0.35→0.80, pose_err inverted 0.25→0, bg_psnr 12→32), weighted mean with garment_sim ×2. VLM overall = mean of six 1–5 criteria.